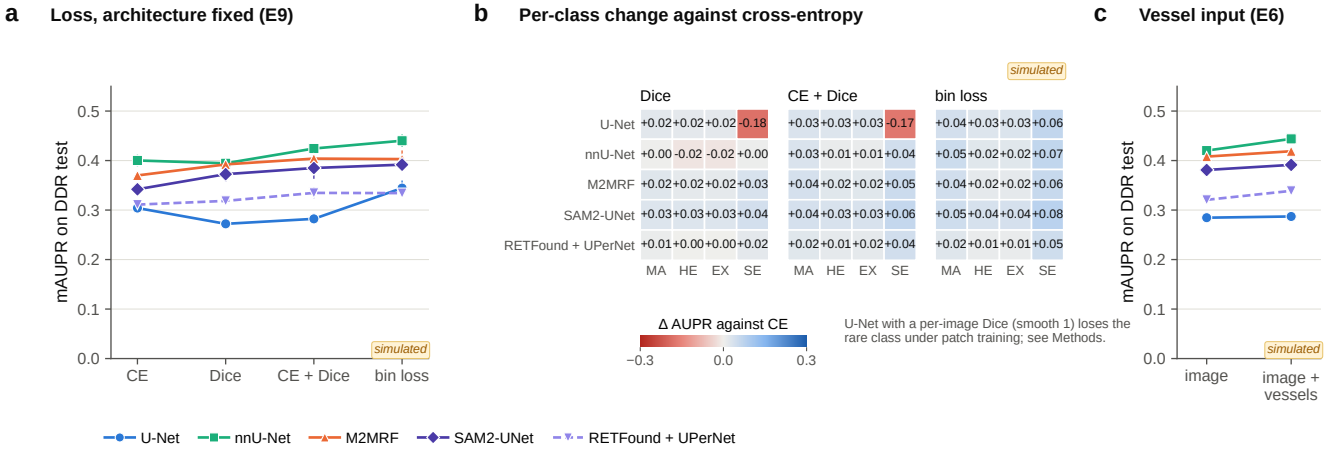
a
Loss, architecture fixed (E9)
mAUPR on DDR test
0.5
0.4
0.3
0.2
0.1
0.0
CE
Dice
CE + Dice
bin loss
simulated
b
Per-class change against cross-entropy
simulated
Dice
CE + Dice
bin loss
U-Net +0.02 +0.02 +0.02 -0.18 +0.03 +0.03 +0.03 -0.17 +0.04 +0.03 +0.03 +0.06
nnU-Net +0.00 -0.02 -0.02 +0.00 +0.03 +0.01 +0.01 +0.04 +0.05 +0.02 +0.02 +0.07
M2MRF +0.02 +0.02 +0.02 +0.03 +0.04 +0.02 +0.02 +0.05 +0.04 +0.02 +0.02 +0.06
SAM2-UNet +0.03 +0.03 +0.03 +0.04 +0.04 +0.03 +0.03 +0.06 +0.05 +0.04 +0.04 +0.08
RETFound + UPerNet +0.01 +0.00 +0.00 +0.02 +0.02 +0.01 +0.02 +0.04 +0.02 +0.01 +0.01 +0.05
MA HE EX SE
Δ AUPR against CE
−0.3
0.0
0.3
U-Net with a per-image Dice (smooth 1) loses the rare class under patch training; see Methods.
c
Vessel input (E6)
mAUPR on DDR test
0.5
0.4
0.3
0.2
0.1
0.0
image
image + vessels
simulated
U-Net
nnU-Net
M2MRF
SAM2-UNet
RETFound + UPerNet